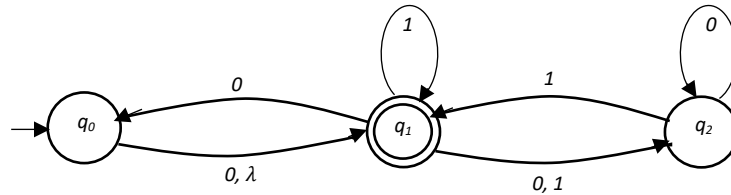


MET CS662 - Assignment #4

Joseph Rissman

U09545917

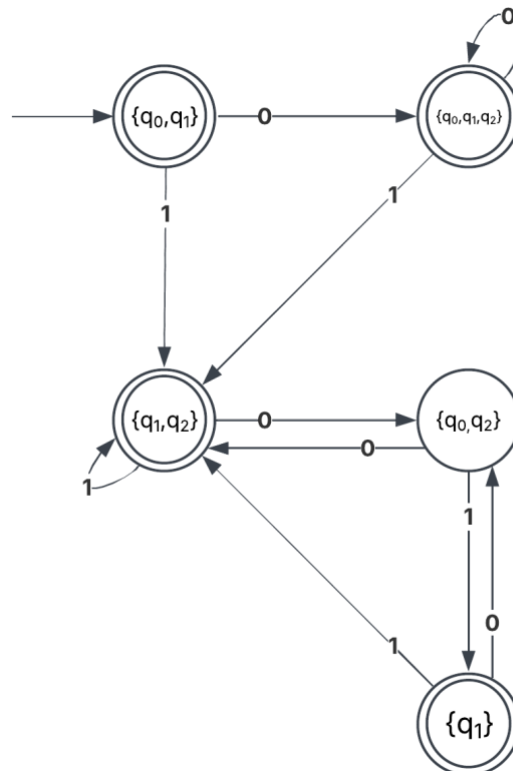
- Convert the following nfa into an equivalent dfa.



Note for TA: The professor didn't specify to use λ -closure or not. I'm guessing from his response that the intended solution is without λ -closure, but I'm submitting both solutions to be safe.

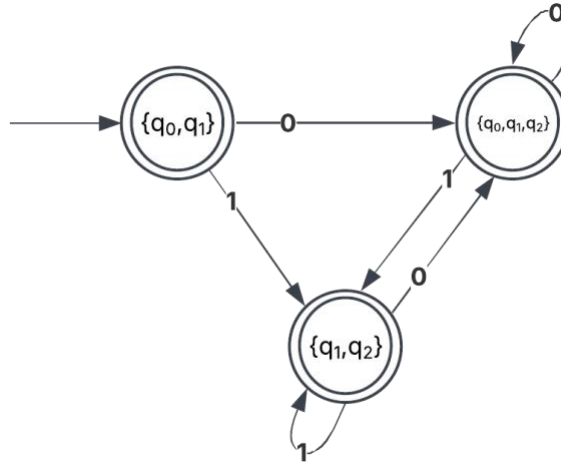
Without λ -closure

	0	1
$\{q_0\}$	$\{q_1\}$	$\emptyset$
$\{q_1\}$	$\{q_0, q_2\}$	$\{q_1, q_2\}$
$\{q_2\}$	$\{q_2\}$	$\{q_1\}$
$\{q_0, q_1\}$	$\{q_0, q_1, q_2\}$	$\{q_1, q_2\}$
$\{q_1, q_2\}$	$\{q_0, q_2\}$	$\{q_1, q_2\}$
$\{q_0, q_2\}$	$\{q_1, q_2\}$	$\{q_1\}$
$\{q_0, q_1, q_2\}$	$\{q_0, q_1, q_2\}$	$\{q_1, q_2\}$

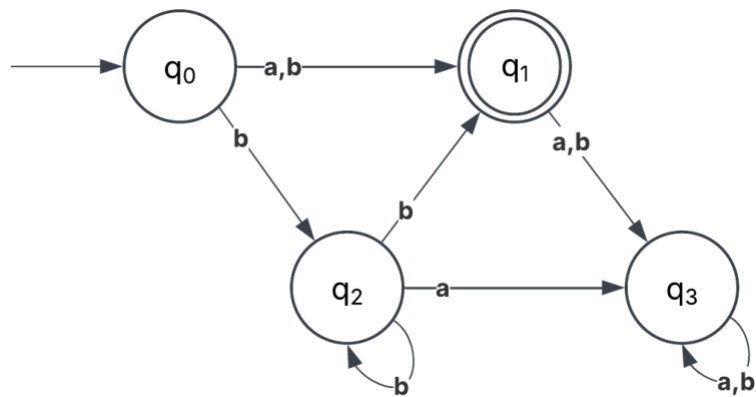


With λ -closure

	λ	0	1
$\{q_0\}$	$\{q_0, q_1\}$	$\{q_1\}$	$\emptyset$
$\{q_1\}$	$\{q_1\}$	$\{q_0, q_1, q_2\}$	$\{q_1, q_2\}$
$\{q_2\}$	$\{q_2\}$	$\{q_2\}$	$\{q_1\}$
$\{q_0, q_1\}$	$\{q_0, q_1\}$	$\{q_0, q_1, q_2\}$	$\{q_1, q_2\}$
$\{q_1, q_2\}$	$\{q_1, q_2\}$	$\{q_0, q_1, q_2\}$	$\{q_1, q_2\}$
$\{q_0, q_1, q_2\}$	$\{q_0, q_1, q_2\}$	$\{q_0, q_1, q_2\}$	$\{q_1, q_2\}$



2. Find an nfa without λ -transitions and with a single final state that accepts the set $\{a\} \cup \{b^n \mid n \geq 1\}$.



3. Find a regular expression for the set $\{a^n b^m \mid (n + m) \text{ is even}\}$

$$(aa)^*(ab)^*(bb)^*$$

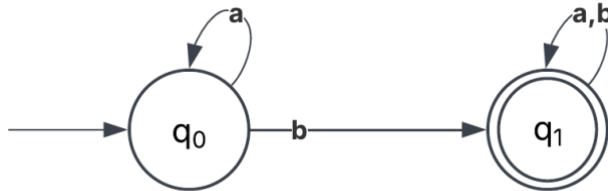
4. Find a regular expression for

$$L = \{\omega \in \{0,1\}^* \mid \omega \text{ has exactly one pair of consecutive zeros}\}$$

(Two 0's in a row appear exactly once: examples 1001011111010111 is OK, 10010001 is not)

$$((0 + \lambda)1)^*(00)(1(0 + \lambda))^*$$

5. Give a nfa that accepts the language $L = ((a + b)^*b(a + bb)^*)$



6. Find all strings in $L = ((a + b)^*b(a + ab)^*)$ of length three or less.

$b, ab, ba, bb, aab, aba, abb, baa, bab, bba, bbb$

7. Find a regular expression for the following language defined on $\{a, b\}$.

$$L = \{\omega \mid n_a(\omega) \text{ and } n_b(\omega) \text{ are both even}\}$$

$$(aa + bb + (ab + ba)(aa + bb)^*(ab + ba))^*$$
